Measure Theoretic Probability
Lesson 7
Extension of Probability Measures: Lemmas 4–6 and the Extension Theorem

Kirschner Awonatey Amaanya

Department of Statistics and Actuarial Science, KNUST

Outline

1. Setting, Tools and Roadmap
2. Lemma 4: The Field Lies Inside $\mathcal{M}$
3. Lemma 5: The Outer Measure Extends $\mathbb{P}$
4. Lemma 6: $\mathbb{P}^*$ Is a Probability Measure on $\mathcal{M}$
5. Completion of the Extension Theorem
6. Summary and Exercises

Outline

1. Setting, Tools and Roadmap
2. Lemma 4: The Field Lies Inside $\mathcal{M}$
3. Lemma 5: The Outer Measure Extends $\mathbb{P}$
4. Lemma 6: $\mathbb{P}^*$ Is a Probability Measure on $\mathcal{M}$
5. Completion of the Extension Theorem
6. Summary and Exercises

The Construction So Far

Setting

Ω is a non-empty set, $\mathcal{F}_0$ a **field** of subsets of Ω, $\mathcal{F} = \sigma(\mathcal{F}_0)$ the generated σ-field, and $\mathbb{P}$ a probability measure on $\mathcal{F}_0$.

Outer measure and the class $\mathcal{M}$

$$\mathbb{P}^*(A) = \inf\Big\{\sum_n \mathbb{P}(A_n) : \ A_n \in \mathcal{F}_0, \ A \subseteq \bigcup_n A_n\Big\}, \qquad A \subseteq \Omega,$$

$$\mathcal{M} = \big\{A \subseteq \Omega : \ \mathbb{P}^*(A \cap B) + \mathbb{P}^*(A^c \cap B) = \mathbb{P}^*(B) \ \ \forall B \subseteq \Omega\big\}.$$

Two More Tools

Properties of $\mathbb{P}$ on the field $\mathcal{F}_0$

(F1) **Finite additivity** for disjoint sets of $\mathcal{F}_0$.

(F2) **Monotonicity**: $A \subseteq B$ in $\mathcal{F}_0$ gives $\mathbb{P}(A) \leq \mathbb{P}(B)$.

(F3) **Countable additivity**: for disjoint $A_n \in \mathcal{F}_0$ with $\bigcup_n A_n \in \mathcal{F}_0$.

The ε-property of the infimum

Since $\mathbb{P}^*(B) \leq 1$ is finite, for every $\varepsilon > 0$ there is a cover $B \subseteq \bigcup_n A_n$ with $A_n \in \mathcal{F}_0$ and

$$\sum_n \mathbb{P}(A_n) < \mathbb{P}^*(B) + \varepsilon.$$

Otherwise $\mathbb{P}^*(B) + \varepsilon$ would be a larger lower bound, contradicting that $\mathbb{P}^*(B)$ is the greatest one.

Field operations

$\mathcal{F}_0$ is closed under complements and finite unions, hence also under finite intersections (De Morgan). So $A, A_n \in \mathcal{F}_0 \Rightarrow A \cap A_n,\ A^c \cap A_n \in \mathcal{F}_0$.

Outline

1. Setting, Tools and Roadmap
2. **Lemma 4: The Field Lies Inside $\mathcal{M}$**
3. Lemma 5: The Outer Measure Extends $\mathbb{P}$
4. Lemma 6: $\mathbb{P}^*$ Is a Probability Measure on $\mathcal{M}$
5. Completion of the Extension Theorem
6. Summary and Exercises

Lemma 4: Statement and Strategy

Lemma 4

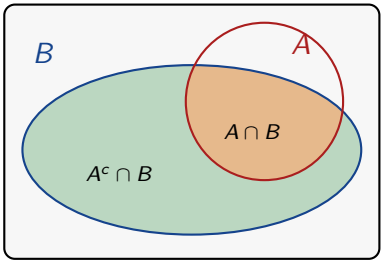
$\mathcal{F}_0 \subseteq \mathcal{M}$.

Strategy

Fix $A \in \mathcal{F}_0$. By the one-sided criterion, it suffices to show, for every $B \subseteq \Omega$,

$$\mathbb{P}^*(A \cap B) + \mathbb{P}^*(A^c \cap B) \leq \mathbb{P}^*(B).$$

Cover B almost optimally by sets of $\mathcal{F}_0$, and **split each covering set** along A.



Proof of Lemma 4 (1): An Almost Optimal Cover of B

Choice of cover

Fix $A \in \mathcal{F}_0$, $B \subseteq \Omega$ and $\varepsilon > 0$. By the ε-property, choose $A_1, A_2, \cdots \in \mathcal{F}_0$ with

$$B \subseteq \bigcup_{n=1}^{\infty} A_n \qquad \text{and} \qquad \sum_{n=1}^{\infty} \mathbb{P}(A_n) < \mathbb{P}^*(B) + \varepsilon.$$

Transfer to $A \cap B$

Intersecting with A and distributing the intersection over the union,

$$A \cap B \ \subseteq \ A \cap \bigcup_{n} A_n = \bigcup_{n=1}^{\infty} (A \cap A_n).$$

Observation

Only the covering property of $\{A_n\}$ has been used so far; nothing special about A. The same holds for A^c: $A^c \cap B \subseteq \bigcup_n (A^c \cap A_n)$.

Proof of Lemma 4 (2): Bounding Each Term

Bound for $\mathbb{P}^*(A \cap B)$

By monotonicity (T2), then subadditivity (T3):

$$\mathbb{P}^*(A \cap B) \;\leq\; \mathbb{P}^*\Big(\bigcup_n (A \cap A_n)\Big) \;\leq\; \sum_{n=1}^{\infty} \mathbb{P}^*(A \cap A_n).$$

The sets $A \cap A_n$ need not be disjoint, but (T3) does not require it.

Passing from $\mathbb{P}^*$ to $\mathbb{P}$

Since $A, A_n \in \mathcal{F}_0$ and $\mathcal{F}_0$ is a field, $A \cap A_n \in \mathcal{F}_0$, so (T4) gives $\mathbb{P}^*(A \cap A_n) \leq \mathbb{P}(A \cap A_n)$. Hence

$$\mathbb{P}^*(A \cap B) \leq \sum_{n=1}^{\infty} \mathbb{P}(A \cap A_n).$$

Same argument for A^c

$A^c \in \mathcal{F}_0$, so likewise $\mathbb{P}^*(A^c \cap B) \leq \sum_{n=1}^{\infty} \mathbb{P}(A^c \cap A_n)$.

Proof of Lemma 4 (3): Conclusion

Adding the two bounds

$$\mathbb{P}^*(A \cap B) + \mathbb{P}^*(A^c \cap B) \;\leq\; \sum_{n=1}^{\infty} \big[\mathbb{P}(A \cap A_n) + \mathbb{P}(A^c \cap A_n)\big].$$

For each n, the sets $A \cap A_n$ and $A^c \cap A_n$ are disjoint elements of $\mathcal{F}_0$ with union A_n. By finite additivity of $\mathbb{P}$ (F1), the bracket equals $\mathbb{P}(A_n)$. Thus

$$\mathbb{P}^*(A \cap B) + \mathbb{P}^*(A^c \cap B) \;\leq\; \sum_{n=1}^{\infty} \mathbb{P}(A_n) \;<\; \mathbb{P}^*(B) + \varepsilon.$$

Let $\varepsilon \downarrow 0$

The inequality is strict for every $\varepsilon > 0$, and the left side does not depend on ε. In the limit it becomes

$$\mathbb{P}^*(A \cap B) + \mathbb{P}^*(A^c \cap B) \leq \mathbb{P}^*(B).$$

As B was arbitrary, $A \in \mathcal{M}$. Hence $\mathcal{F}_0 \subseteq \mathcal{M}$. □

Outline

1. Setting, Tools and Roadmap
2. Lemma 4: The Field Lies Inside $\mathcal{M}$
3. **Lemma 5: The Outer Measure Extends $\mathbb{P}$**
4. Lemma 6: $\mathbb{P}^*$ Is a Probability Measure on $\mathcal{M}$
5. Completion of the Extension Theorem
6. Summary and Exercises

Lemma 5: Statement and Plan

Lemma 5

For every $A \in \mathcal{F}_0$,

$$\mathbb{P}^*(A) = \mathbb{P}(A).$$

Plan

(T4) already gives $\mathbb{P}^*(A) \leq \mathbb{P}(A)$. It remains to prove $\mathbb{P}(A) \leq \mathbb{P}^*(A)$, i.e. that $\mathbb{P}(A)$ is a **lower bound** for every covering sum $\sum_n \mathbb{P}(A_n)$. Since $\mathbb{P}^*(A)$ is the *greatest* lower bound, this gives $\mathbb{P}(A) \leq \mathbb{P}^*(A)$.

Proof of Lemma 5: A Lower Bound for All Covering Sums

Setup

Let $A \in \mathcal{F}_0$ and $A_1, A_2, \cdots \in \mathcal{F}_0$ with $A \subseteq \bigcup_n A_n$. Then $A = \bigcup_n (A \cap A_n)$. We want $\mathbb{P}(A) \leq \sum_n \mathbb{P}(A_n)$.

Why not $\mathbb{P}(A) \leq \mathbb{P}(\bigcup_n A_n) \leq \sum_n \mathbb{P}(A_n)$?

$\mathbb{P}$ is defined only on $\mathcal{F}_0$, and $\mathcal{F}_0$ is a field: the countable union $\bigcup_n A_n$ need not belong to $\mathcal{F}_0$, so $\mathbb{P}(\bigcup_n A_n)$ may be undefined.

Disjointification inside $\mathcal{F}_0$

Put $B_n = A \cap A_n \cap A_1^c \cap \cdots \cap A_{n-1}^c$. Then $B_n \in \mathcal{F}_0$, $B_n \subseteq A_n$, the B_n are pairwise disjoint, and $\bigcup_n B_n = A \in \mathcal{F}_0$.

Proof of Lemma 5: Conclusion

Computation

The disjoint union $\bigcup_n B_n = A$ lies in $\mathcal{F}_0$, so (F3) applies; then monotonicity (F2) on each term:

$$\mathbb{P}(A) = \sum_{n=1}^{\infty} \mathbb{P}(B_n) \;\leq\; \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

Taking the infimum

This holds for **every** cover $\{A_n\}$ of A by sets of $\mathcal{F}_0$. So $\mathbb{P}(A)$ is a lower bound of the set of covering sums, and $\mathbb{P}^*(A)$ is its greatest lower bound:

$$\mathbb{P}(A) \leq \mathbb{P}^*(A).$$

Conclusion

Together with (T4), $\mathbb{P}^*(A) = \mathbb{P}(A)$ for all $A \in \mathcal{F}_0$. In particular $\mathbb{P}^*(\Omega) = \mathbb{P}(\Omega) = 1$. □

Outline

1. Setting, Tools and Roadmap
2. Lemma 4: The Field Lies Inside $\mathcal{M}$
3. Lemma 5: The Outer Measure Extends $\mathbb{P}$
4. **Lemma 6: $\mathbb{P}^*$ Is a Probability Measure on $\mathcal{M}$**
5. Completion of the Extension Theorem
6. Summary and Exercises

Lemma 6

Lemma 6

$\mathbb{P}^*$ is a probability measure on the measurable space $(\Omega, \mathcal{M})$.

Proof

By Lemma 3, $(\Omega, \mathcal{M})$ is a measurable space. Three requirements remain:

1. **Null set:** $\mathbb{P}^*(\emptyset) = 0$, by (T1).
2. **Countable additivity:** for pairwise disjoint $A_1, A_2, \cdots \in \mathcal{M}$, Lemma 2(b) with test set $B = \Omega$ gives $\mathbb{P}^*\left(\bigcup_n A_n\right) = \sum_n \mathbb{P}^*(A_n)$.
3. **Total mass:** $\Omega \in \mathcal{F}_0$, so by Lemma 5, $\mathbb{P}^*(\Omega) = \mathbb{P}(\Omega) = 1$.

Values lie in $[0, 1]$ by monotonicity: $0 \leq \mathbb{P}^*(A) \leq \mathbb{P}^*(\Omega) = 1$. □

Remark

Items 1 and 2 make $\mathbb{P}^*$ a **measure**; item 3 makes it a **probability** measure. Countable additivity cannot come from (T3) alone (that is only an inequality). It is Lemma 2 that supplies equality.

Outline

1. Setting, Tools and Roadmap
2. Lemma 4: The Field Lies Inside $\mathcal{M}$
3. Lemma 5: The Outer Measure Extends $\mathbb{P}$
4. Lemma 6: $\mathbb{P}^*$ Is a Probability Measure on $\mathcal{M}$
5. **Completion of the Extension Theorem**
6. Summary and Exercises

The Generated σ-Field Lies Inside $\mathcal{M}$

Claim

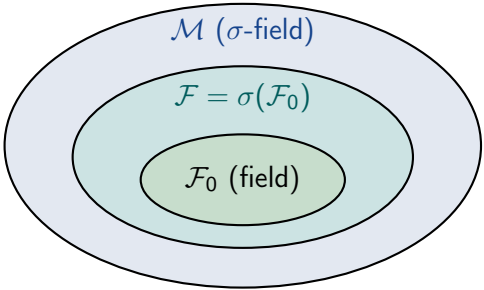
$\mathcal{F}_0 \subseteq \mathcal{F} = \sigma(\mathcal{F}_0) \subseteq \mathcal{M}$.

Proof

The first inclusion is the definition of $\mathcal{F}$. By Lemma 3, $\mathcal{M}$ is a σ-field, and by Lemma 4 it contains $\mathcal{F}_0$. Since $\mathcal{F}$ is the **smallest** σ-field containing $\mathcal{F}_0$, we get $\mathcal{F} \subseteq \mathcal{M}$.

Equivalent argument (Lesson 2)

If $\mathcal{F}$ stuck out of $\mathcal{M}$, then $\mathcal{F} \cap \mathcal{M}$ would be a σ-field containing $\mathcal{F}_0$ and smaller than $\mathcal{F}$: a contradiction.



The Extension Theorem

Theorem (extension of a probability measure from a field)

Let $\mathbb{P}$ be a probability measure on a field $\mathcal{F}_0$ and $\mathcal{F} = \sigma(\mathcal{F}_0)$. Then the restriction of $\mathbb{P}^*$ to $\mathcal{F}$ is a probability measure on $(\Omega, \mathcal{F})$ and

$$\mathbb{P}^*(A) = \mathbb{P}(A) \qquad \text{for every } A \in \mathcal{F}_0.$$

Proof

Since $\mathcal{F} \subseteq \mathcal{M}$, $\mathbb{P}^*$ is defined on $\mathcal{F}$, and every statement below concerns sets of $\mathcal{F} \subseteq \mathcal{M}$.

1. $\emptyset \in \mathcal{F}$ and $\mathbb{P}^*(\emptyset) = 0$.
2. If $A_n \in \mathcal{F}$ are pairwise disjoint, then $\bigcup_n A_n \in \mathcal{F}$ (as $\mathcal{F}$ is a σ-field) and $A_n \in \mathcal{M}$, so $\mathbb{P}^*(\bigcup_n A_n) = \sum_n \mathbb{P}^*(A_n)$ by Lemma 6.
3. $\Omega \in \mathcal{F}_0 \subseteq \mathcal{F}$ and $\mathbb{P}^*(\Omega) = 1$.

Agreement with $\mathbb{P}$ on $\mathcal{F}_0$ is Lemma 5. □

Uniqueness of the Extension (Recalled)

Theorem

If $\mathbb{Q}$ is any probability measure on $\mathcal{F}$ with $\mathbb{Q} = \mathbb{P}$ on $\mathcal{F}_0$, then $\mathbb{Q} = \mathbb{P}^*$ on $\mathcal{F}$.

Idea of the proof (Lesson 5)

- For a cover $A \subseteq \bigcup_n A_n$ with $A_n \in \mathcal{F}_0$: $\mathbb{Q}(A) \leq \sum_n \mathbb{Q}(A_n) = \sum_n \mathbb{P}(A_n)$, so $\mathbb{Q}(A) \leq \mathbb{P}^*(A)$ (greatest lower bound).
- Apply this to $A^c \in \mathcal{F}$ and use the complement rule for the two probability measures $\mathbb{Q}$ and $\mathbb{P}^*$: $\mathbb{P}^*(A) \leq \mathbb{Q}(A)$.

Conclusion

A probability measure on a field has **exactly one** extension to the generated σ-field.

Outline

1. Setting, Tools and Roadmap
2. Lemma 4: The Field Lies Inside $\mathcal{M}$
3. Lemma 5: The Outer Measure Extends $\mathbb{P}$
4. Lemma 6: $\mathbb{P}^*$ Is a Probability Measure on $\mathcal{M}$
5. Completion of the Extension Theorem
6. **Summary and Exercises**

Summary of Lemmas 4–6

Results of this lesson

Lemma	Statement	Main ingredients
4	$\mathcal{F}_0 \subseteq \mathcal{M}$	ε-cover of B; (T2)–(T4); finite additivity of $\mathbb{P}$; $\varepsilon \to 0$
5	$\mathbb{P}^* = \mathbb{P}$ on $\mathcal{F}_0$	disjointification in $\mathcal{F}_0$; (F3); greatest lower bound
6	$\mathbb{P}^*$ is a probability measure on $\mathcal{M}$	(T1); Lemma 2 with $B = \Omega$; Lemma 5

Consequence

Every probability measure specified on a **field** extends, in exactly one way, to the σ-field it generates.

Exercises

1. Let $\mathcal{F}_0 = \{\emptyset, \Omega\}$ with $\mathbb{P}(\emptyset) = 0$, $\mathbb{P}(\Omega) = 1$. Compute $\mathbb{P}^*(A)$ for every $A \subseteq \Omega$ and determine $\mathcal{M}$.
2. Explain precisely where the hypothesis "$\mathcal{F}_0$ is closed under finite intersections" is used in the proof of Lemma 4.
3. Show that a finite subfamily of the proof of Lemma 5 does not need (F3), and explain why the countable case does.
4. Using the extension theorem, describe a probability measure on $\sigma(\mathcal{F}_0)$ for the field $\mathcal{F}_0$ of finite disjoint unions of intervals $(a, b] \cap [0, 1]$ given by $\mathbb{P}((a, b]) = b - a$.

Outlook

Next lesson: limits of sets

Up to now, families such as $A_1, A_2, A_3, \ldots$ were **collections** of sets whose order played no role. We now study **sequences** of sets, where the order is meaningful, and ask what the limit of a sequence of sets could mean. The tools are the **limit infimum** and the **limit supremum** of a sequence of sets.

Motivation

For a sequence of numbers the limit is another number. For a sequence of sets, one expects the limit to be another set.